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333653(33)

BE (6th Semester)
Examination, April - May, 2016

(New Scheme)

Software Engineering and Project Management

Time Allowed : 3 hours

Maximum Marks : 80
Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of every question is compulsory. Solve any two from parts (*b*), (*c*) and (*d*). Draw diagram wherever required.

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) Define software engineering. 2

(*b*) Explain spiral model with its advantages and disadvantages. 7

(*c*) Explain at least ten software quality and productivity factors. 7

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(*d*) What are the managerial issues related to software development? Discuss some of them with their solution?. 7

2. (*a*) Define requirement engineering. 2

(*b*) What do you mean by requirement elicitation, and what are the various techniques to do this process? 7

(*c*) What are the major factors that influence software cost? 7

(*d*) Explain SCM process and its need in software engineering. 7

3. (*a*) What is walk through? 2

(*b*) What are the various notations that are used to specify the external characteristics, architectural structure and procesing details of a software system? 7

(*c*) Explain the types of coupling and cohesion. 7

(*d*) What are the various kinds of activities prescribed in a test plan? 7

4. (*a*) What is mutation testing? 2

(*b*) What are the various levels or stages of testing a software? 7

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(c) For a given flow graph, determine :

(i) The cyclomatic complexity

(ii) The independent paths

(iii) Number of regions

(iv) Number of edges

(v) Number of flow graph nodes

(vi) Number of predicate nodes

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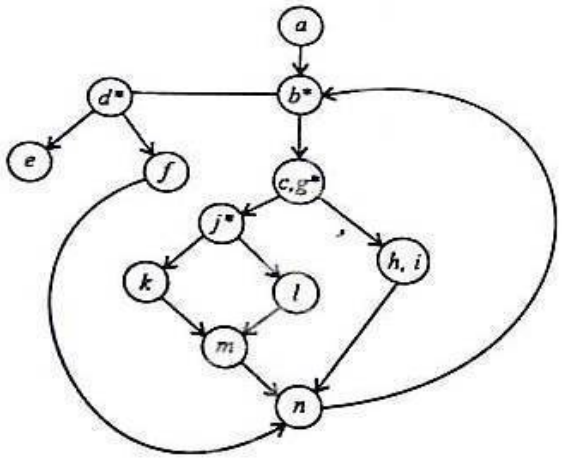

(d) Explain debugging and various debugging techniques. 7

5. (a) Define CASE. 2

(b) What are the various software cost estimation techniques used in software engineering? Explain in brief. 7

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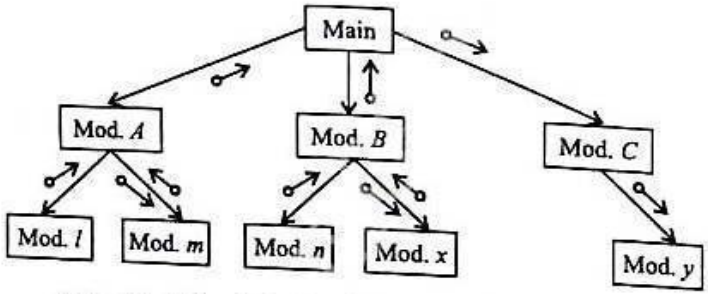
(c) What is information flow metrics? Calculate the information flow index of individual modules as well as whole software for the given structure chart.



(d) Explain COCOMO model. For a product size of 200 KLOC to be developed. Software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate effort, development time, average staff size and productivity of the project. ($a=3.0$, $b=1.12$, $c=2.5$ and $d=0.35$). 7

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